$$0.7660 = \frac{AC + 8}{20 + 10.44} \Rightarrow AC = 23.32 - 8 \ = 15.32 \text{ m}$$

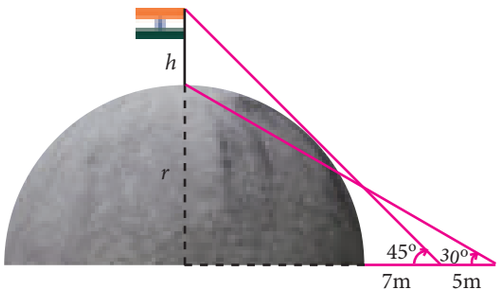
Therefore, the horizontal distance between two trees $= AC = 15.32$ m

Thinking Corner

1. What type of triangle is used to calculate heights and distances?
2. When the height of the building and distances from the foot of the building is given, which trigonometric ratio is used to find the angle of elevation?
3. If the line of sight and angle of elevation is given, then which trigonometric ratio is used
   (i) to find the height of the building
   (ii) to find the distance from the foot of the building.

Exercise 6.2

1. Find the angle of elevation of the top of a tower from a point on the ground, which is 30 m away from the foot of a tower of height $10\sqrt{3}$ m.
2. A road is flanked on either side by continuous rows of houses of height $4\sqrt{3}$ m with no space in between them. A pedestrian is standing on the median of the road facing a row house. The angle of elevation from the pedestrian to the top of the house is $30°$. Find the width of the road.
3. To a man standing outside his house, the angles of elevation of the top and bottom of a window are 60° and 45° respectively. If the height of the man is 180 cm and if he is 5 m away from the wall, what is the height of the window? $(\sqrt{3} = 1.732)$
4. A statue 1.6 m tall stands on the top of a pedestal. From a point on the ground, the angle of elevation of the top of the statue is 60° and from the same point the angle of elevation of the top of the pedestal is 40°. Find the height of the pedestal. $(\tan 40° = 0.8391,\ \sqrt{3} = 1.732)$
5. A flag pole of height 'h' metres is on the top of the hemispherical dome of radius 'r' metres. A man is standing 7 m away from the dome. Seeing the top of the pole at an angle $45°$ and moving 5 m away from the dome and seeing the bottom of the pole at an angle $30°$. Find (i) the height of the pole (ii) radius of the dome. $(\sqrt{3} = 1.732)$



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